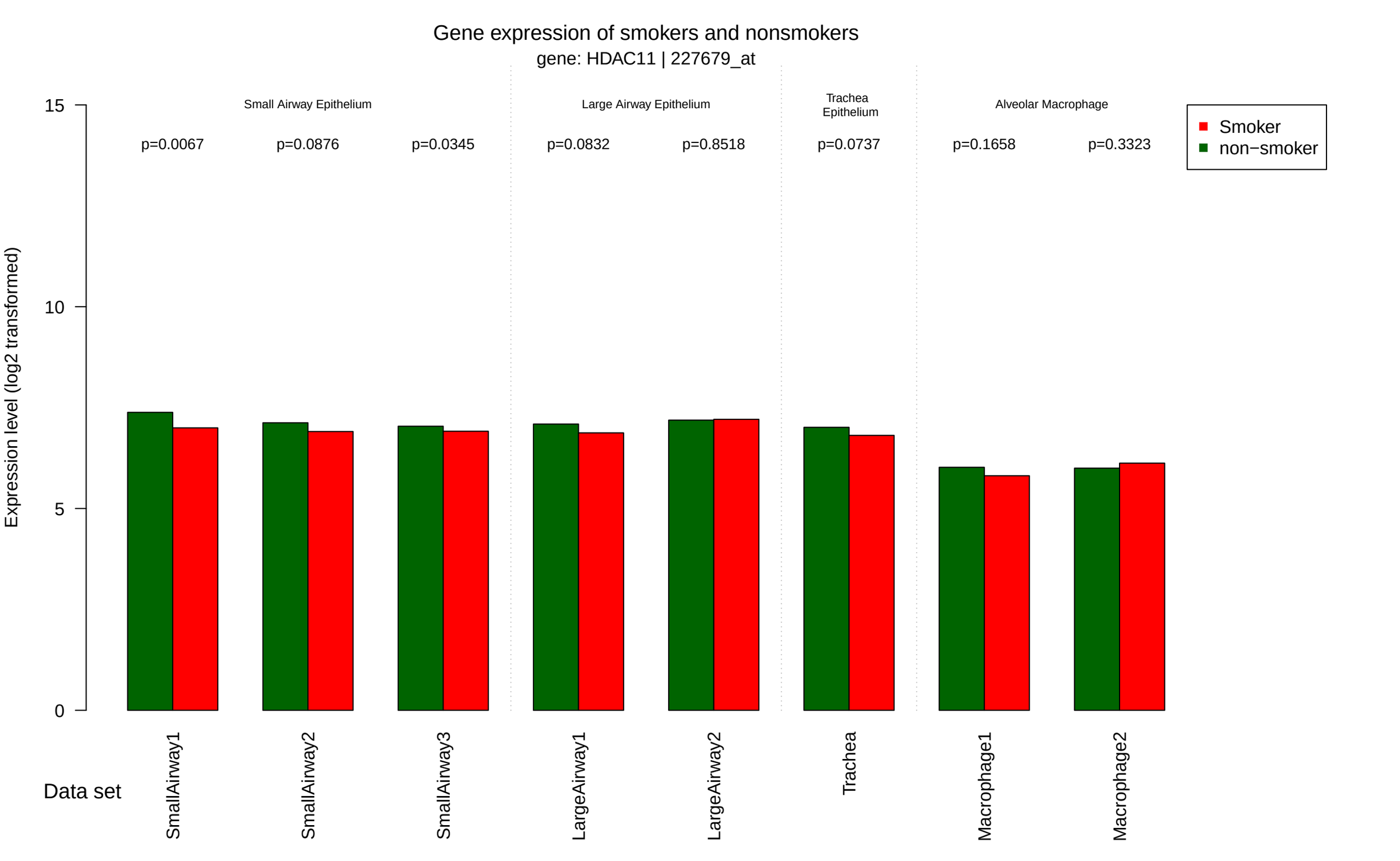
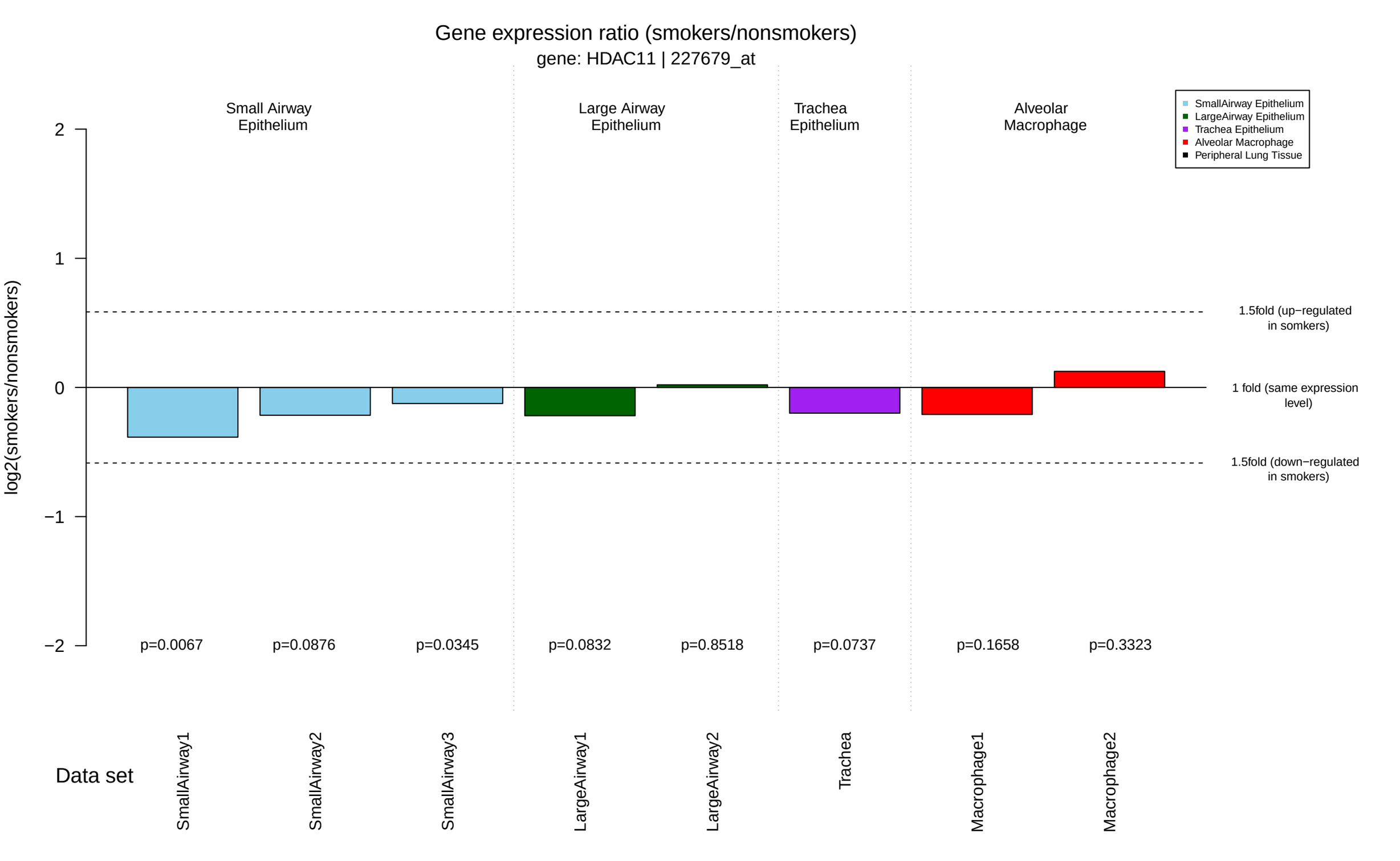
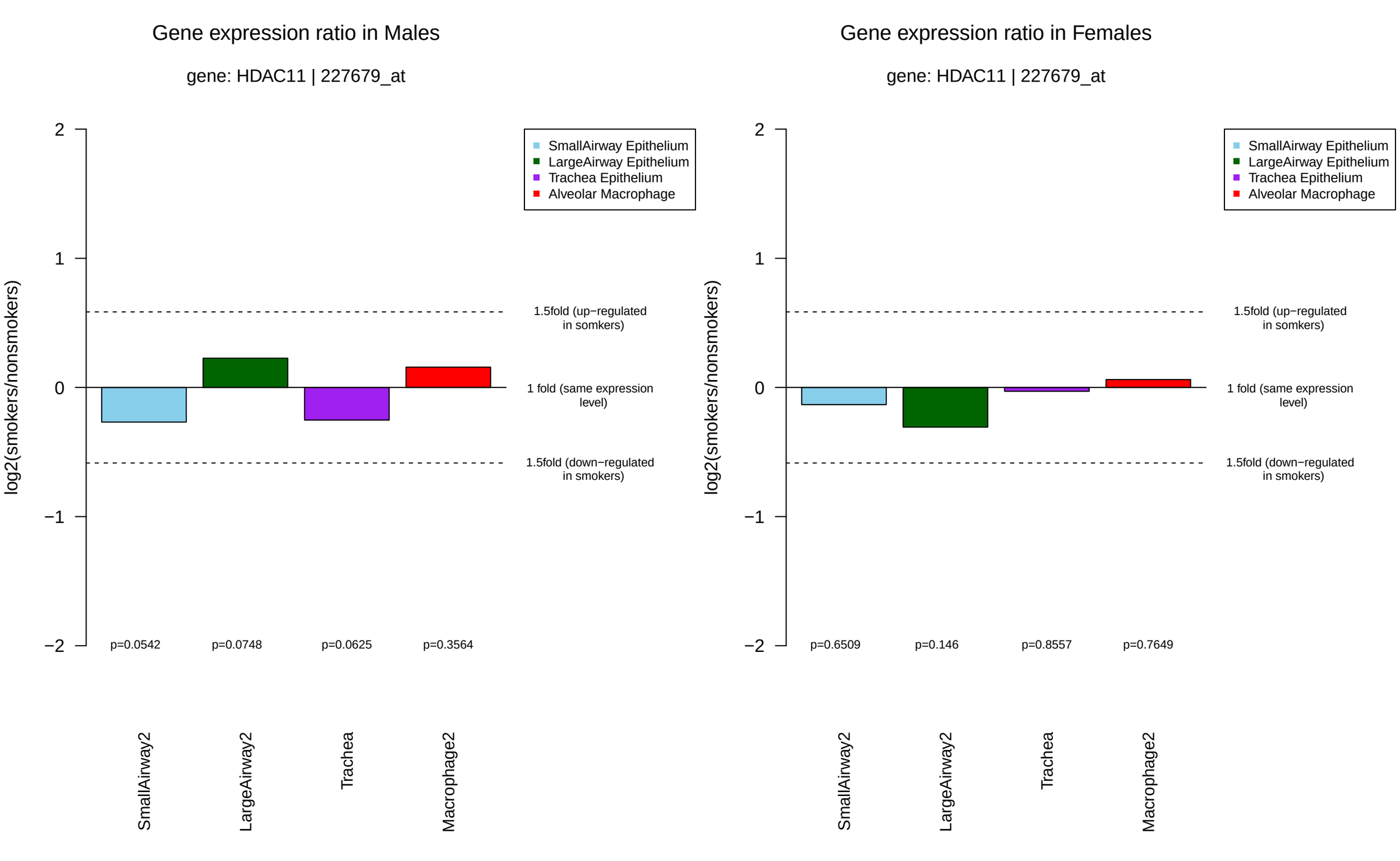
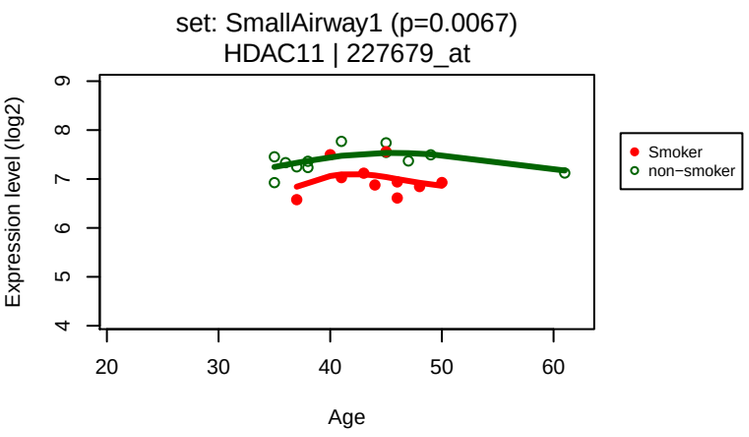
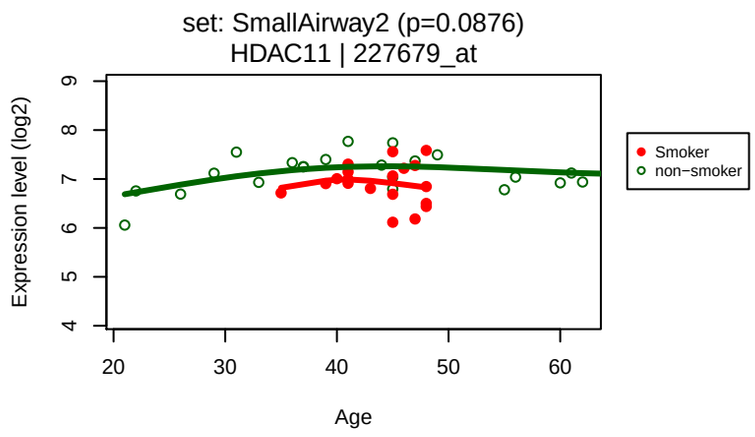
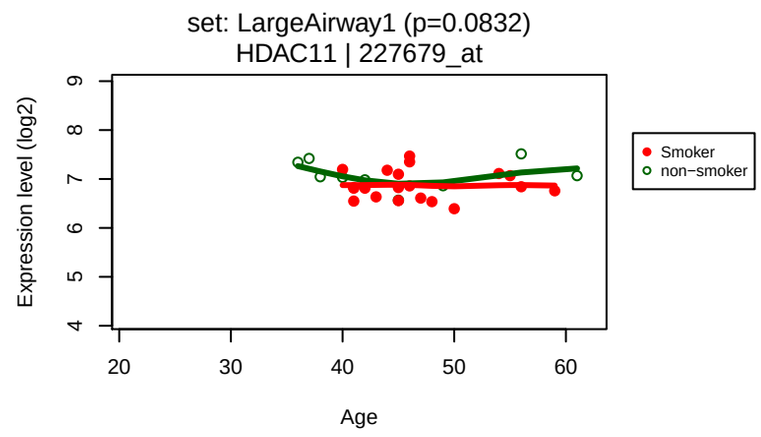

Gene expression of smokers and nonsmokers
gene: HDAC11 | 227679_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0067
p=0.0876
p=0.0345
p=0.0832
p=0.8518
p=0.0737
p=0.1658
p=0.3323
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

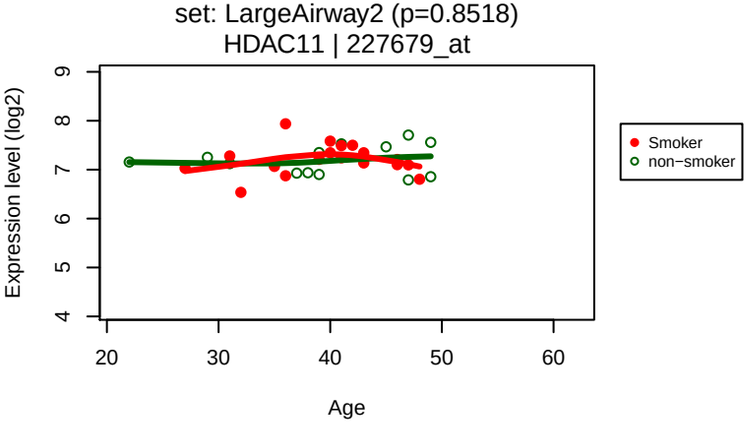

Gene expression ratio (smokers/nonsmokers)
gene: HDAC11 | 227679_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0067
p=0.0876
p=0.0345
p=0.0832
p=0.8518
p=0.0737
p=0.1658
p=0.3323
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

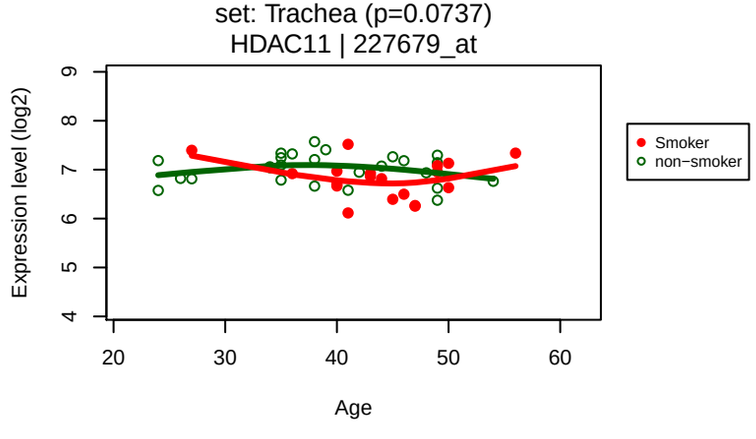

Gene expression ratio in Males
gene: HDAC11 | 227679_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0542
p=0.0748
p=0.0625
p=0.3564
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HDAC11 | 227679_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6509
p=0.146
p=0.8557
p=0.7649
SmallAirway2
LargeAirway2
Trachea
Macrophage2

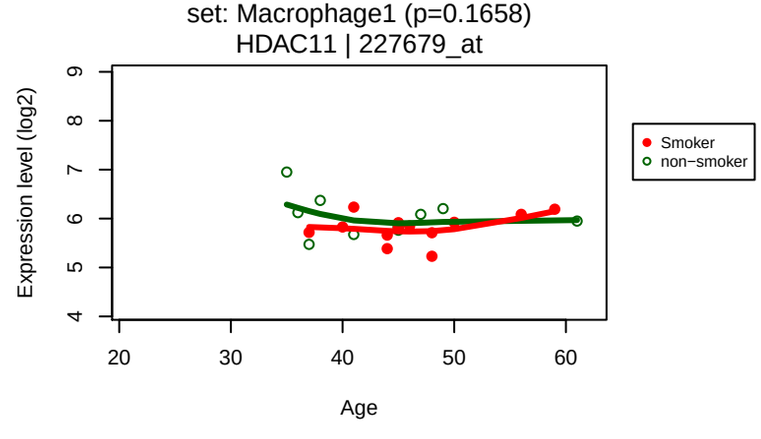

set: SmallAirway1 (p=0.0067)
HDAC11 | 227679_at
Expression level (log2)
Age
Smoker
non−smoker

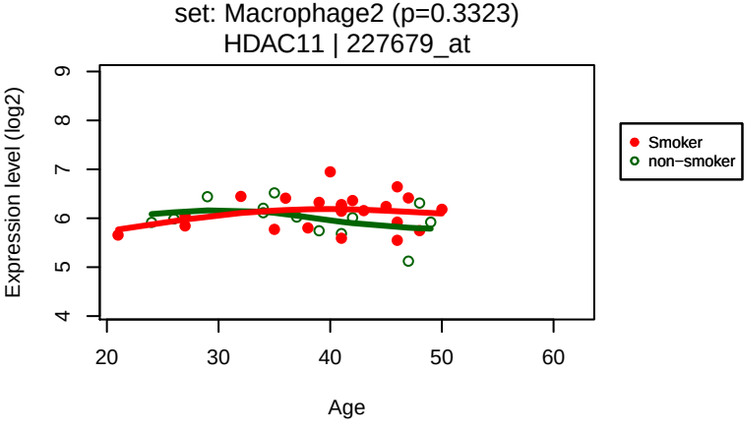

set: SmallAirway2 (p=0.0876)
HDAC11 | 227679_at
Expression level (log2)
Age
Smoker
non−smoker

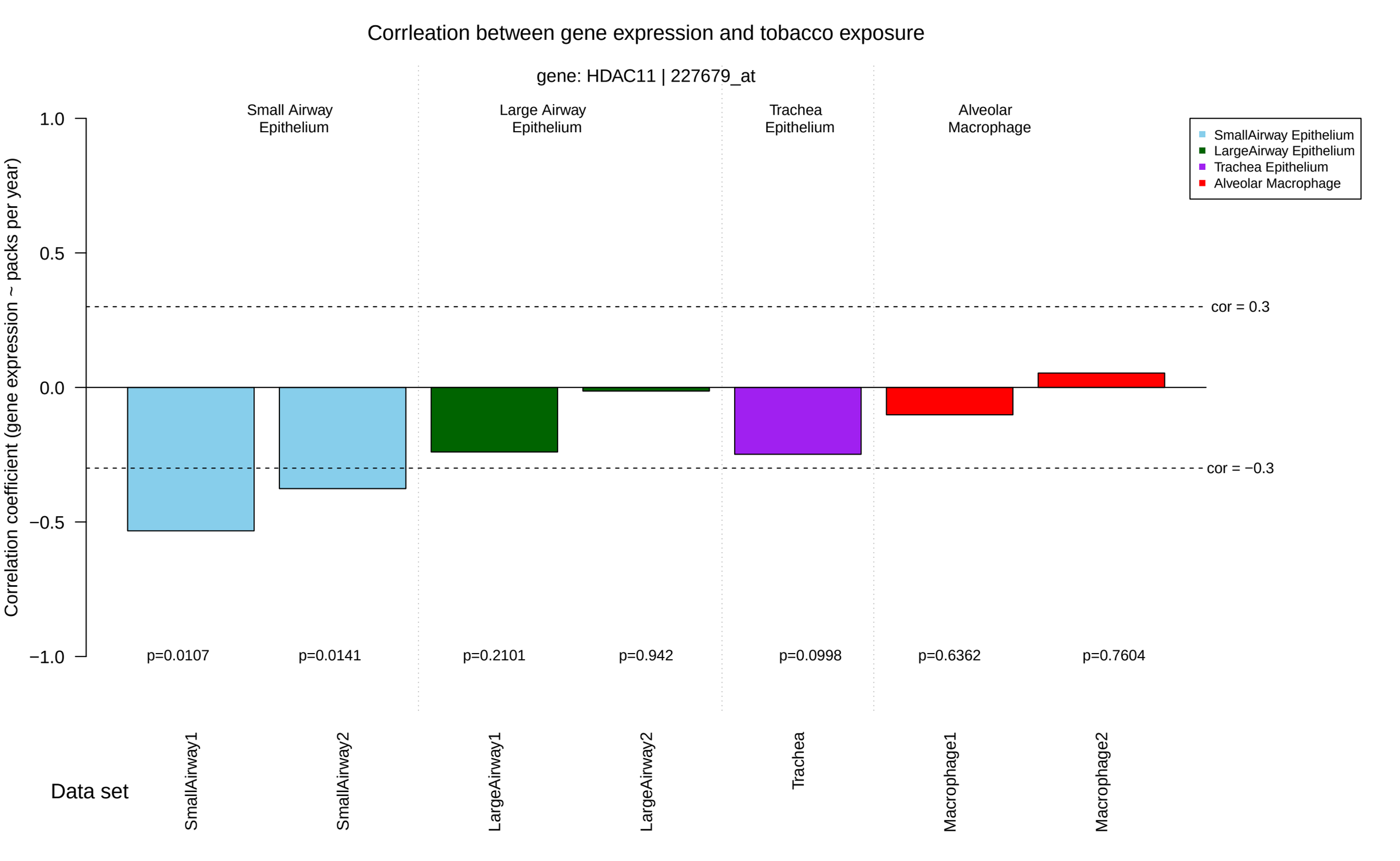

set: LargeAirway1 (p=0.0832)
HDAC11 | 227679_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8518)
HDAC11 | 227679_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0737)
HDAC11 | 227679_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1658)
HDAC11 | 227679_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3323)
HDAC11 | 227679_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HDAC11 | 227679_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0107
p=0.0141
p=0.2101
p=0.942
p=0.0998
p=0.6362
p=0.7604
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2